

MARTIN MILON

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Summary:

M.S. Data Science (May 2026, Summa Cum Laude) with 2+ years building machine learning systems; career transitioner seeking entry-level data science or analytics engineering roles where an economics background or domain expertise in legal operations is an edge.

Education:

M.S. Data Science | New York Institute of Technology | Graduation Date: May 2026

B.A. Economics & Political Science | Stony Brook University | Graduation Date: December 2020

Projects:

Reorder Prediction | <https://github.com/martinmmv/Instacart-Reorder-Prediction>

Supervised Machine Learning | May 2026

- ❖ Built an end-to-end binary classification pipeline on 3M+ real grocery orders from 200K+ Instacart users and 50K+ products to predict whether a customer will reorder a product in their next basket.
- ❖ Trained and tuned three ML models—Logistic Regression, Decision Tree, and LightGBM—with 5-fold cross-validation and threshold optimization; the best model (LightGBM, ROC-AUC = 0.971, F1 = 0.953) outperformed the Logistic Regression baseline by 0.21 percentage points on ROC-AUC.
- ❖ Deployed the model in three stages: (1) a prediction function; (2) a batch scoring pipeline; and (3) a Flask REST API.

Customer Churn | <https://github.com/martinmmv/Customer-Churn>

Supervised Machine Learning | May 2026

- ❖ Built an end-to-end binary classification pipeline following the CRISP-DM framework to predict customer churn using IBM's Telco dataset (+7,000 instances).
- ❖ Implemented and tuned four ML algorithms—ZeroR, Decision Tree, Naïve Bayes, SVM—with 10-fold cross-validation; the best model (SVM, F1 = 0.84) outperformed the ZeroR baseline model by 22 percentage points.
- ❖ Applied AdaBoost and Bagging ensemble methods and developed a real-world deployment plan covering model monitoring, retraining cadence, and data pipeline maintenance.

YouTube Data Pipeline | <https://github.com/martinmmv/YouTube-Data-Pipeline>

Data Engineering | In Progress

- ❖ Designing and building an end-to-end ETL pipeline in Python that collects daily snapshots of channel and video engagement data (views, likes, comments, subscriber counts) from the YouTube Data API v3 across multiple channels, storing longitudinal records in a local PostgreSQL database for time-series analysis.
- ❖ Engineered a quota-efficient data collection layer, reducing API quota consumption by ~98%; implemented a quota tracker class that monitors daily usage (10,000-unit limit), logs warnings at 80% consumption, and halts execution before quota exhaustion.
- ❖ Designed a normalized relational schema in PostgreSQL conforming to Boyce-Codd Normal Form, separating static entity metadata (channels, videos) from time-varying measurements.
- ❖ Built a two-layer data validation system: input validation applied before API calls to prevent quota waste, and record-level validation applied after collection and before database insertion.
- ❖ Planning exploratory data analysis, time-series engagement trend analysis, and content categorization across channels once the data collection layer is complete.

Professional Experience:

Paralegal | McBean Law, PLLC | White Plains, New York

June 2025 - February 2026 | Full-time

- ❖ Oversaw a diverse immigration caseload involving multiple benefit types before the U.S. Citizenship and Immigration Services, including but not limited to naturalizations, adjustments of status, immigrant visas, and waivers.
- ❖ Caseload also included multiple forms of relief and protection before the immigration courts and the Board of Immigration Appeals, including applications for asylum and withholding of removal, as well as motions to reopen and dismiss proceedings.
- ❖ Built a relational case management model and dashboard in Microsoft Excel to analyze cases by type, status, jurisdiction, assigned attorney, time elapsed, and outcome; track multiple deadlines per case; track the status and details of hearings for cases with pending removal proceedings, and track the delivery status and duration of document submissions.

Senior Caseworker | Cherokee Federal | Manhattan, New York

March 2024 - June 2025 | Full-time

- ❖ Oversaw a complex humanitarian caseload at the Humanitarian Emergency Response and Relief Center (HERRC) located at the Roosevelt Hotel in Midtown Manhattan.
- ❖ Responsible for collecting and validating case management information, preparing reports on caseload and services, and collaborating with government agencies and nonprofit organizations.
- ❖ Built an optimization model to improve the delivery of case management services and resource allocation under case-specific legal, demographic, and medical constraints; as well as team-based case management strategies under multiple constraints, including HERRC dynamic policies, limited resources, the number of caseworkers per shift, and shift duration.

Paralegal | Law Office of Alexandra Mayen Rivera, P.C. | Huntington, New York

May 2022 - January 2024 | Full-time

- ❖ Oversaw a diverse immigration caseload involving multiple forms of relief and protection before the immigration courts and the Board of Immigration Appeals, including applications for asylum and for withholding of removal, non-LPR cancellation of removal, and motions to reopen proceedings.
- ❖ Oversaw the firm's Freedom of Information Act (FOIA) caseload, including records from U.S. Citizenship and Immigration Services, Customs and Border Protection, and the Office of Biometric Identity Management, and requests for records of proceedings from the immigration courts.
- ❖ Oversaw cases involving prosecutorial discretion requests before the U.S. Department of Homeland Security.

Awards:

Summa Cum Laude | New York Institute of Technology | May 2026 | Graduated with Distinction

Community Service Award | Stony Brook University | May 2018 | EOP/AIM

Certificate of Appreciation | Stony Brook University | August 2017 | Physics Teaching Assistant

Outstanding Academic Achievement Award | Stony Brook University | August 2016 | Physics

Skills:

Python | Pandas | NumPy | Matplotlib | Seaborn | SciKitLearn

Other | R | SQL | Excel | Visual Studio Code | Google Colab | Databricks | Weka | D3.js

Bilingual Professional | English | Spanish